

ENAE601 Astrodynamics

Monday, August 31, 2026 9:27 PM

- ELMS
- Some figures shown in class can be downloaded from go.umd.edu/ENAE601-figures

Law of universal gravitation - two-body motion

Two point masses orbiting about their center of mass

One mass is much greater than the other, barycenter = cm of that body

Kepler's laws

1. Orbit of a planet is an ellipse, with the sun at the focus
2. Line from the sun to planet sweeps out equal areas in equal times
3. For any two planets

$$\frac{p_1^2}{a_1^3} = \frac{p_2^2}{a_2^3}$$

Newton

$$\vec{F} = - \frac{GMm}{r^2} \hat{r}$$

$\hat{r}$ ← unit vector
distance from earth center to satellite

$$GM = \mu = 398600 \frac{\text{km}^3}{\text{s}^2}$$

1. Independent of velocity and time
2. A central force, i.e., directed along the radial line connecting the bodies
3. Inverse square dependence on distance

1. Conservation of energy
2. Conservation of angular momentum
3. Conservation of eccentricity vector

$$m \ddot{\vec{r}} = \vec{F} = - \frac{\mu m}{r^2} \hat{r}$$

$$m \ddot{\vec{r}} = \vec{F} = -\frac{\mu}{r^2} \hat{r}$$

$$\ddot{\vec{r}} + \frac{\mu}{r^2} \hat{r} = 0$$

Equation of motion

$$\vec{F} = -\vec{\nabla} V(\vec{r}) \Rightarrow \ddot{\vec{r}} + \vec{\nabla} V(\vec{r}) = 0$$

$$\dot{\vec{r}} \cdot \dot{\vec{r}} = \dot{\vec{r}} \cdot \vec{\nabla} V(\vec{r}) = 0$$

$$\frac{d}{dt} \frac{\dot{\vec{r}} \cdot \dot{\vec{r}}}{2} + \frac{d}{dt} V(\vec{r}) = 0$$

$$\frac{d}{dt} \left(\frac{\dot{\vec{r}} \cdot \dot{\vec{r}}}{2} + V(\vec{r}) \right) = 0$$

$\frac{1}{2} v^2$ $\xrightarrow{\quad}$ $\underbrace{\hspace{1cm}}_{\text{energy}} \xleftarrow{\quad}$ $V(\vec{r})$ potential energy

$$\mathcal{E} = \frac{v^2}{2} - \frac{\mu}{r} \quad V(\vec{r}) = -\frac{\mu}{r}$$

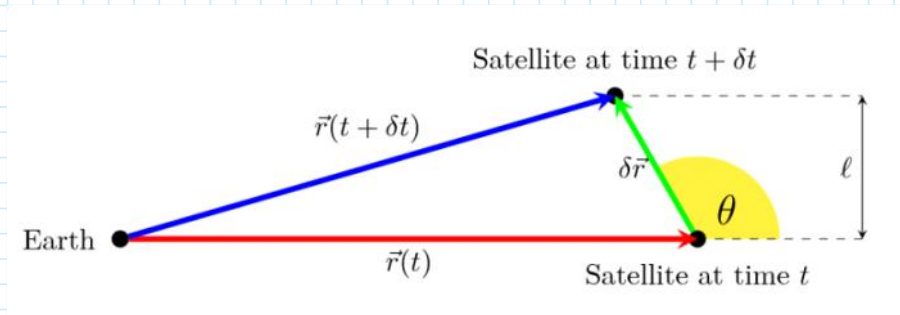
Angular momentum

Curtis section 2.4

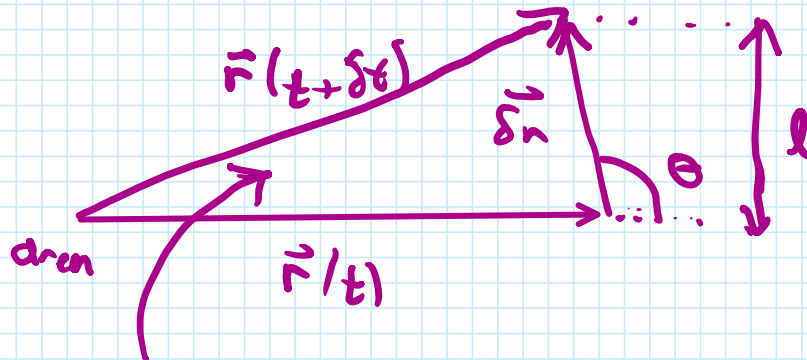
$$\vec{F} \propto \hat{r}, \quad \ddot{\vec{r}} \propto \vec{F}$$

$$\underbrace{\vec{r} \times \ddot{\vec{r}}}_{\text{angular momentum}} \propto \vec{r} \times \vec{r} = 0$$

$$\frac{d}{dt} \underbrace{(\vec{r} \times \dot{\vec{r}})}_{\vec{h} \text{ angular momentum}} = 0$$



<https://go.umd.edu/ENAE601-figures>



$$\delta A = \frac{1}{2} \delta r \sin \theta = \frac{1}{2} |\vec{r} \times \delta \vec{r}|$$

$$\begin{aligned} \dot{A} &= \lim_{\delta t \rightarrow 0} \frac{\delta A}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{1}{2} \frac{|\vec{r} \times \delta \vec{r}|}{\delta t} \\ &= \frac{1}{2} \underbrace{|\vec{r} \times \dot{\vec{r}}|}_{\vec{h}} = \frac{h}{2} \end{aligned}$$

Area swept out over a finite time interval from t_1 to t_2 is

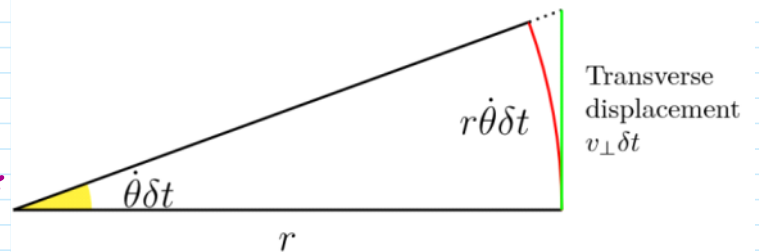
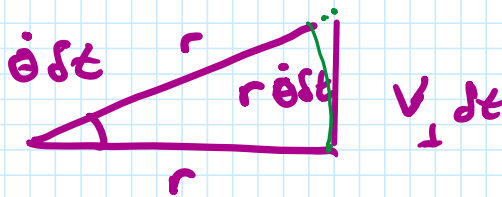
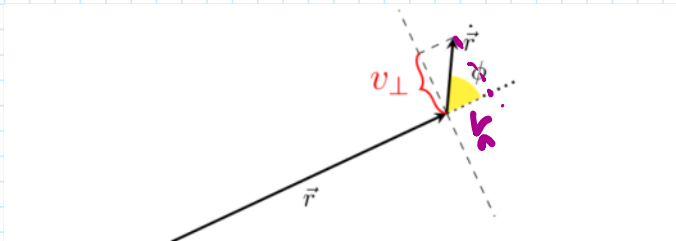
$$A_{t_1, t_2} = \int_{t_1}^{t_2} \frac{1}{2} h dt = \frac{1}{2} h (t_2 - t_1)$$

Kepler's second law proved.

Transverse velocity



Curtis figure 2.9



$$v_{\perp} = r \dot{\theta} \quad \text{in limit } \delta t \rightarrow 0$$

$$h = r v \sin \phi = r v_{\perp} = r^2 \dot{\theta}$$

Mathematical interlude

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} (\vec{A} \cdot \vec{C}) - \vec{C} (\vec{A} \cdot \vec{B})$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$

$$\dot{r} = \frac{d}{dt} |\vec{r}|$$

|| ? No!

$$v = \left| \frac{d}{dt} \vec{r} \right|$$

$$\dot{r} = \frac{d}{dt} |\vec{r}| = \frac{d}{dt} \sqrt{x^2 + y^2 + z^2}$$

$$= \frac{\frac{dx}{dt}x + \frac{dy}{dt}y + \frac{dz}{dt}z}{\sqrt{x^2 + y^2 + z^2}} = \frac{\vec{r} \cdot \dot{\vec{r}}}{r}$$

$$r \dot{r} = \vec{r} \cdot \dot{\vec{r}}$$

This is a true relationship that looks like it ought to be true

... to be continued

Lecture 2
9/3/2026

$$\begin{aligned} \dot{\vec{r}} \times \vec{h} &= - \frac{1}{r^3} \vec{r} \times \vec{h} \\ &= \frac{d}{dt} (\vec{r} \times \vec{h}) \\ &= - \vec{r} \times \dot{\vec{h}} = - \vec{r} \times (\vec{r} \times \dot{\vec{r}}) \end{aligned}$$

$$\frac{1}{r^3} \vec{r} \times \vec{h} = \frac{1}{r^3} \vec{r} \times \vec{h}$$

$$\begin{aligned}
 -\vec{r} \times \dot{\vec{h}} &= -\dot{\vec{r}} \times (\vec{r} \times \dot{\vec{r}}) \\
 &= -\dot{\vec{r}}(\vec{r} \cdot \dot{\vec{r}}) + \dot{\vec{r}}(\dot{\vec{r}} \cdot \vec{r}) \\
 &= r^2 \dot{\vec{r}} - r \dot{r} \vec{r}
 \end{aligned}$$

$$\begin{aligned}
 \mu \frac{d}{dt} \hat{r} &= \mu \frac{d}{dt} \left(\frac{\vec{r}}{r} \right) = \frac{\mu}{r} \dot{\vec{r}} - \frac{\mu}{r^2} \dot{r} \vec{r} \\
 &= -\frac{\mu}{r^3} (\vec{r} \times \dot{\vec{h}})
 \end{aligned}$$

$$\frac{d}{dt} (\vec{r} \times \dot{\vec{h}}) = \mu \frac{d}{dt} \hat{r}$$

$$\frac{d}{dt} \underbrace{\left(\frac{\vec{r} \times \dot{\vec{h}}}{\mu} - \hat{r} \right)}_{\vec{e}} = 0 \quad \vec{A} = \mu \vec{e}$$

Orbital motion in space from conservation laws

$$\vec{e} + \hat{r} = \frac{\vec{r} \times \dot{\vec{h}}}{\mu}$$

$$\underbrace{\vec{r} \cdot \vec{e}}_{r \cos \theta} + r = \frac{1}{\mu} \vec{r} \cdot (\dot{\vec{r}} \times \vec{h}) = \frac{1}{\mu} \underbrace{(\vec{r} \times \dot{\vec{r}})}_{\vec{h}} \cdot \vec{h} = \frac{h^2}{\mu}$$

$$r(1 + e \cos \theta) = \frac{h^2}{\mu} = p \quad \text{Semi-latus rectum}$$

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Eccentricity $e = |\vec{e}|$

Terminology: "anomaly", "argument" = angle

The orbit equation

Curtis (2.45)

$$r = \frac{p}{1 + e \cos \theta}$$

Relates r , geocentric distance, to θ true anomaly

+x aligned with direction of eccentricity vector

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$r = p - ex$$

$$\begin{aligned} y^2 &= r^2 - x^2 = (p - ex)^2 - x^2 \\ &= p^2 - 2epx + \underbrace{e^2 x^2 - x^2}_{-(1-e^2)x^2} \\ &= p^2 - 2epx - (1-e^2)x^2 \end{aligned}$$

Define $a = \frac{p}{1-e^2}$

Subst. $p = a(1-e^2)$

$$\begin{aligned} y^2 &= ap(1-e^2) - 2ae(1-e^2)x - (1-e^2)x^2 \\ &= (1-e^2)(ap - 2aex - x^2) \end{aligned}$$

$$\begin{aligned}
 &= (1-e^2)(a_p - 2ae x - x^2) \\
 &= (1-e^2) \left[\underbrace{a^2 e^2 + a_p}_{a^2} - (ae+x)^2 \right] \\
 &= (1-e^2) [a^2 - (ae+x)^2]
 \end{aligned}$$

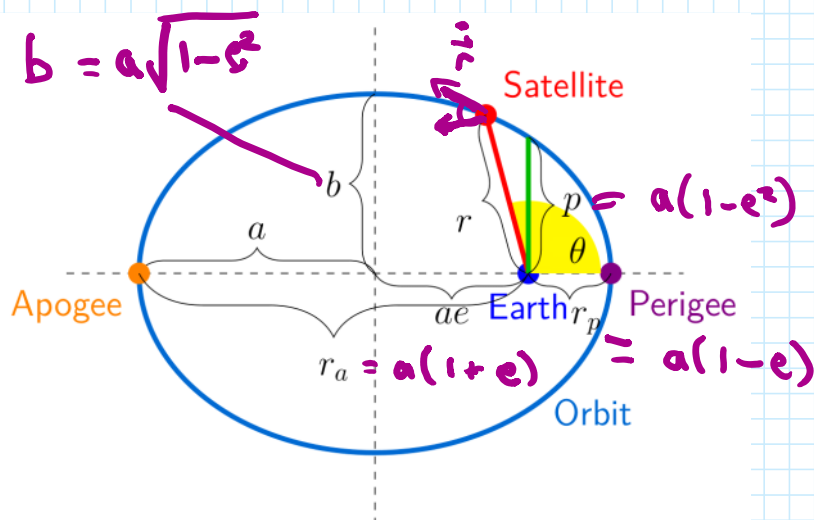
$$\frac{(x+ae)^2}{a^2} + \frac{y^2}{a^2(1-e^2)} = 1 \quad \text{Equation of conic section}$$

Circle $e=0$ of radius a

Ellipse: $1 > e > 0$ a is the semimajor axis

Parabola: $e=1$

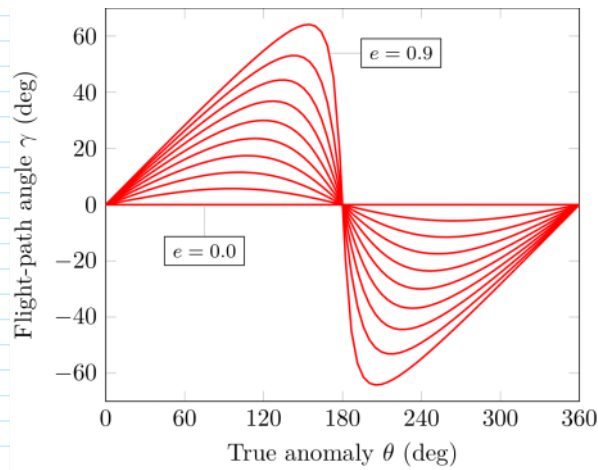
Hyperbola: $e > 1$



Flight-path angle

Curtis section 2.4, pages 73-74, figure 2.12

Angle from local horizontal to the velocity vector



$$\tan \gamma = \frac{v_r}{v_t} = \frac{e \sin \theta}{1 + e \cos \theta}$$

Relation of physical and geometric quantities

$$e^2 = \frac{1}{\mu^2} (\dot{\vec{r}} \times \vec{h}) \cdot (\dot{\vec{r}} \times \vec{h}) - \frac{2}{\mu r} \underbrace{\vec{r} \cdot (\dot{\vec{r}} \times \vec{h})}_{h^2} + 1$$

$$|\dot{\vec{r}} \times \vec{h}| = v h$$

$$e^2 = \frac{v^2 h^2}{\mu^2} - \frac{2 h^2}{\mu r} + 1$$

$$1 - e^2 = p \left(\frac{2}{r} - \frac{v^2}{\mu} \right)$$

$$p = \left(\frac{1 - e^2}{\left(\frac{2}{r} - \frac{v^2}{\mu} \right)} \right) \Rightarrow a = \left(\frac{2}{r} - \frac{v^2}{\mu} \right)^{-1}$$

$$\epsilon = \frac{v^2}{2} - \frac{\mu}{r}$$

$$= - \frac{\mu}{2a}$$

$$a = \left(- \frac{2}{\mu} \epsilon \right)^{-1}$$

Summary of orbital motion so far

- Orbit is in a plane
 - a, semimajor axis, size
 - e, eccentricity, shape
 - True anomaly a function of time ??? —
- It is a conic section

Perifocal coordinates

$\hat{p}$ = perigee

$\hat{q}$ = in orbital plane $\perp \hat{p}$

$\hat{w} = \hat{h}$